

A man with a beard, wearing a light-colored suit and tie, is seated in the driver's seat of a car. He is looking out the window with a serious expression. The interior of the car is dimly lit, while the background outside is dark. The overall mood is somber and intense.

CHICAGO CRIME ANALYSIS

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ABOUT THE PROJECT

This project aimed to analyze crime trends in the city of Chicago using publicly available crime data. With over 8 million crime records, we performed stratified sampling, explored key crime patterns, validated two hypotheses statistically and visually, and built a dynamic Tableau dashboard for stakeholder decision-making.



OBJECTIVE

To identify temporal and geographic crime patterns, evaluate assumptions about crime behavior, and derive actionable insights for public safety initiatives and resource allocation.

Chicago Crime Dataset Overview

[DATASET LINK](#)

- **Source:** Published by the City of Chicago, collected by the Chicago Police Department
- **Time Span:** Covers reported crimes from 2001 to present
- **Update Frequency:** Updated daily
- **Contents:**
 - Type of crime
 - Location and time of occurrence
 - Whether an arrest was made
- **Data Collection:**
 - Derived from official police reports
 - Cleaned and organized before public release
- **Reliability:**
 - Considered trustworthy and reliable due to its official origin and regular updates
- **Limitations:**
 - Only includes reported crimes
 - Some incidents may be missing or underreported
- **Usefulness:**

Valuable for analyzing crime trends, patterns, and geographic distribution in Chicago

PROBLEMS IN DATASET

LARGE DATASET SIZE

Full dataset caused memory overload and slow performance in tools like Excel or Tableau.

IRRELEVANT COLUMNS

Columns like IUCR, FBI Code, and Case Number were not useful and added unnecessary bulk.

MISSING & INCONSISTENT DATA

Fields like Location Description had null values.

Category names had inconsistencies (e.g., "STREET" vs. "Street") impacting grouping

IMBALANCED DATA DISTRIBUTION

Some crime types or years had very few records, making balanced sampling tricky.

STRATIFIED SAMPLING



- To create a representative subset, stratified sampling was applied to the Chicago Crime Dataset.
- A new column called Strata was formed by combining Year and Primary Type, grouping crimes by both time and category.
- A 10% sample was extracted using `train_test_split` with the `stratify` parameter, ensuring each subgroup was proportionally represented.
- This method preserved the overall distribution of crime types across years, making the subset suitable for reliable analysis.

CHI-SQUARE TEST OF INDEPENDENCE

FOR ARREST COLUMN

WHY USED: This test is suitable for comparing two categorical distributions — in this case, the distribution of Arrest values (True/False) between the full dataset and the sample.

FORMULA: $\chi^2 = \sum((O_i - E_i)^2 / E_i)$

WHERE: O_i = Observed frequency
 E_i = Expected frequency under the assumption of independence

RESULT: P-value > 0.05

THIS INDICATES THERE IS NO SIGNIFICANT DIFFERENCE IN ARREST DISTRIBUTIONS BETWEEN THE SAMPLE AND FULL DATASET.



KOLMOGOROV-SMIRNOV (KS) TEST

FOR HOUR COLUMN

WHY USED:

The KS test compares the distributions of two continuous (numerical) variables — here, the distribution of crime hours in the sample vs. full dataset.

FORMULA:

$$D = \sup_x |F_1(x) - F_2(x)|$$

WHERE:

$F_1(x)$, $F_2(x)$ are the empirical cumulative distribution functions (ECDF) of the two datasets.

D is the maximum distance between the ECDFs.

RESULT:

P-value > 0.05

THIS MEANS THE HOUR-WISE DISTRIBUTION OF CRIMES IS NOT SIGNIFICANTLY DIFFERENT BETWEEN THE FULL DATA AND THE SAMPLE.



STATISTICAL TEST RESULTS SUMMARY

Feature	Test Used	Test Statistic	P-Value & Interpretation
Arrest	Chi-Square Test	0.2484	0.6182 → No significant difference. Sample is valid.
Domestic	Chi-Square Test	0.3295	0.5660 → Proportionally similar. Sample is valid.
Location Description	Chi-Square Test	213.4843	0.5356 → No significant difference. Sample is valid.
Community Area	Chi-Square Test	59.0559	0.9360 → Distribution consistent. Sample is good.
Hour	KS Test	0.0006	0.9538 → Distribution preserved. Sample is representative.

HANDLING NULL VALUES



- Community Area was imputed using latitude and longitude via a Chicago Data Portal reference file.
- For missing latitude or longitude, the average location within the same community area was used.
- These steps ensured the dataset remained clean and suitable for analysis.
- Location Description was treated as essential; records missing it were removed
- Overall, only 0.2% of the sampled data was dropped due to missing Location Description.

FINAL DATA SET AFTER REMOVING UNWANTED ATTRIBUTES

DATA DIMENSIONS

- ORIGINAL SIZE: 782,597 ROWS × 12 COLUMNS (USED FOR DASHBOARD)
- PROCESSED SIZE: 404,925 ROWS × 12 COLUMNS (48% REDUCTION)(USED FOR HYPOTHESIS TESTING)
- TIME SPAN: 22+ YEARS (2001-2023)
- NO MISSING VALUES IN ANY COLUMN AFTER PREPROCESSING

Column	Data Type	Description	Key Insights
ID	Integer	Unique crime record identifier	Range: 1.3M to 13.1M
Date	Datetime	Crime occurrence timestamp	Full temporal coverage
Primary Type	Categorical	Main crime category	Multiple crime types available
Description	Categorical	Detailed crime subcategory	Granular crime classification
Location Description	Categorical	Crime location type	Various venue types
Arrest	Boolean	Arrest made (True/False)	Binary outcome variable
Domestic	Boolean	Domestic-related incident	Binary flag
Community Area	Integer	Chicago community area (0-77)	Geographic segmentation
Year	Integer	Crime year	2001-2023 range
Latitude	Float	Geographic coordinate	36.6° to 42.0° range
Longitude	Float	Geographic coordinate	-91.7° to -87.5° range
Hour	Integer	Hour of occurrence (0-23)	Temporal granularity

HYPOTHESIS FORMULATION

TESTING WHETHER THE DISTRIBUTION OF CRIME TYPES (PRIMARY TYPE) VARIES BY LOCATION (LOCATION DESCRIPTION)

- Null Hypothesis (H_0): The distribution of crime types is independent of location type; i.e., crime type occurs equally regardless of location.
- Alternative Hypothesis (H_1): The distribution of crime types depends on the location type; i.e., certain crimes are more/less likely at certain locations.

TESTING WHETHER THE DISTRIBUTION OF CRIME TYPE CHANGES BY TIME (E.G., HOUR OF THE DAY)”

- Null hypothesis (H_0): The distribution of crime types does NOT change by hour; time and crime type are independent.
- Alternative hypothesis (H_1): The distribution of crime types DOES change by hour; time and crime type are associated.

HYPOTHESIS TESTING #1

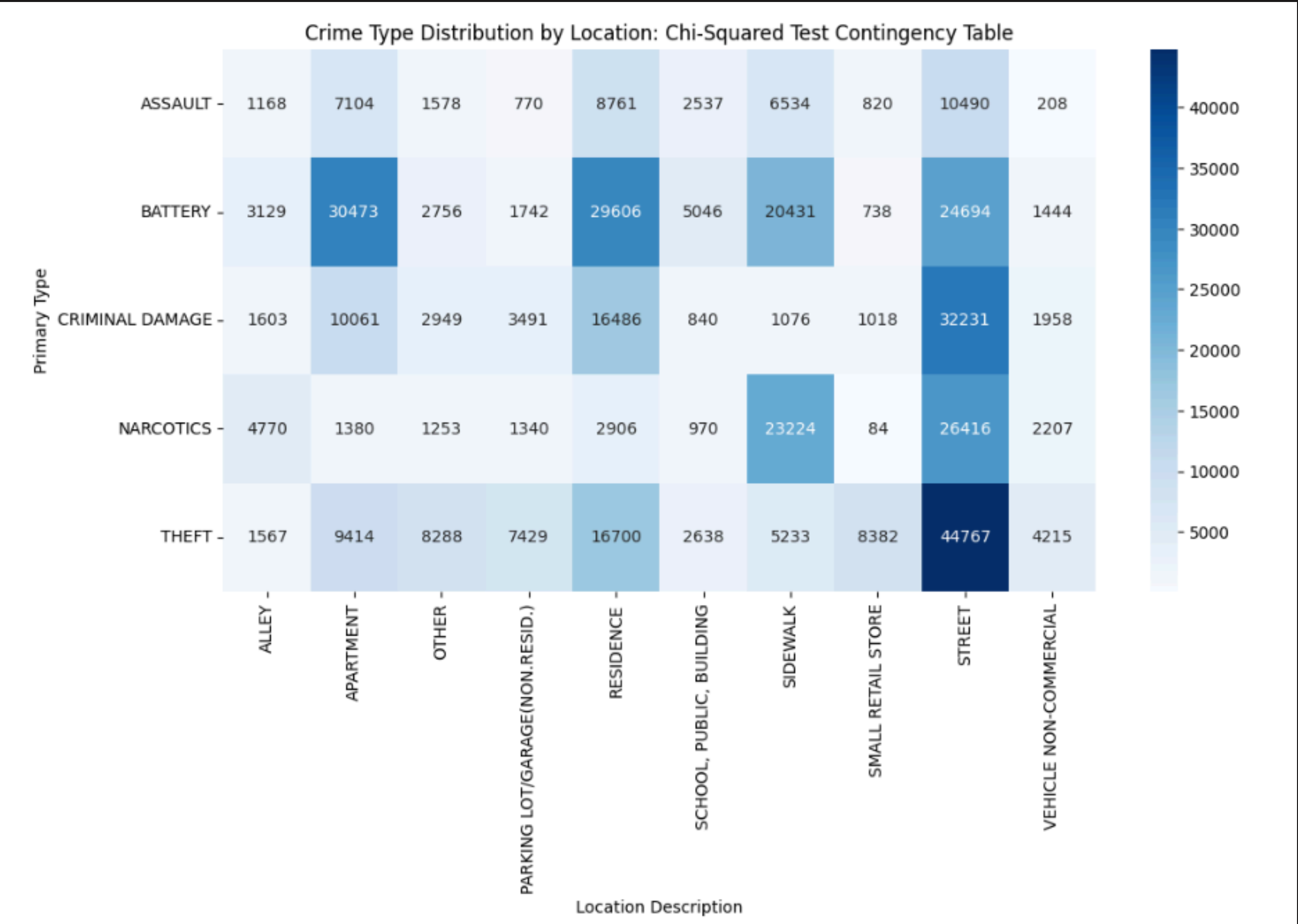
CRIME TYPE VS LOCATION

HYPOTHESIS: Crime type distribution changes by location

RESULT: Chi-square = 116,296.41
p-value $\approx$ 0.0
degree of freedom = 36

CONCLUSION: Strong evidence that crime types vary significantly by location

CRIME TYPE VS LOCATION



HYPOTHESIS TESTING #2

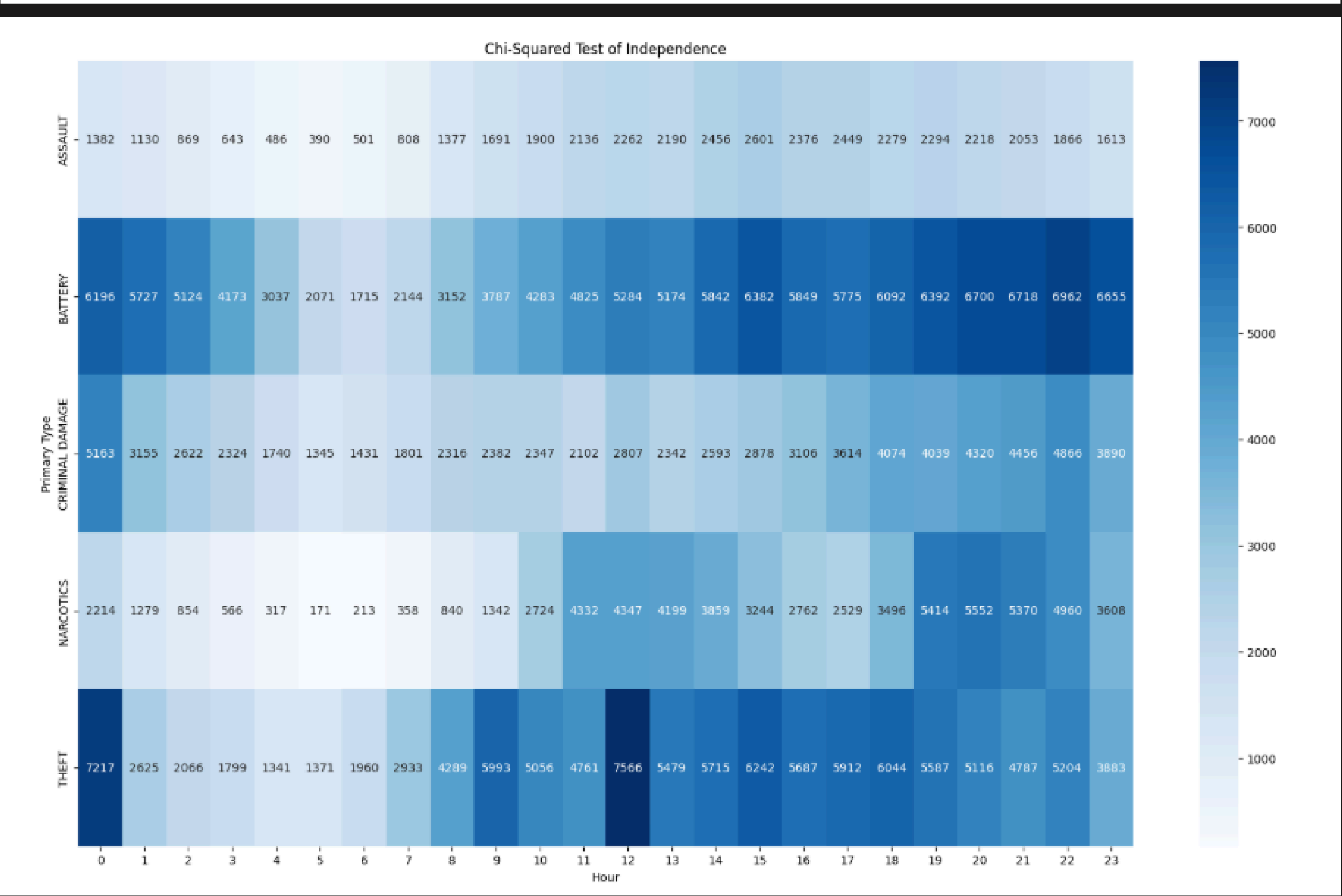
CRIME TYPE VS TIME

HYPOTHESIS: Crime type distribution changes by hour of day

RESULT: Chi-square = 23245.97
p-value $\approx$ 0.0
degree of freedom = 92

CONCLUSION: Strong evidence that crime types vary significantly by time (hour of day)

CRIME TYPE VS TIME



HANDING OVER THE DASHBOARD TO YOU



THANK YOU